Computer Workshop Practice

Ground to Mastery — Study Notes

Tools, Safety, Maintenance, Printers, Scanners, E-Waste

ICT 121 | Year 1 Semester 2 | DCS

Coursework 40% (Test 10% + Assignment 10% + Practical 20%) | Final Exam 60%

How to Study This Course

This is a hands-on practical course. Theory explains what you do in the lab. Study by understanding WHY each procedure is done, not just memorising steps. Exam questions test both knowledge and procedure.

> 📝 *Assessment: Test 10% + Assignment 10% + Practical 20% + Exam 60%. Practical component means you will be assessed doing real tasks.*

UNIT 1: LABORATORY TOOLS & SAFETY

Workshop tools, safety gadgets, first aid, fire safety

▸ 1.1 — Computer Workshop Tools

Term / Concept	Definition / Explanation
`Flathead screwdriver`	Turns slotted screws — various sizes needed
`Phillips (cross) screwdriver`	Turns cross-head screws — most common in computers
`Torx screwdriver`	Star-shaped — used in newer laptops and Macs
`Needle-nose pliers`	Grips small components, picks up jumpers and small screws
`ESD wrist strap`	Electrostatic Discharge protection — clips to ground — prevents static damage to components
`Anti-static mat`	Work surface that dissipates static electricity
`Multimeter`	Measures voltage, current, resistance — tests PSU, cables
`Thermal paste`	Heat-conducting compound between CPU and heatsink
`POST diagnostic card`	Plugs into motherboard — shows error codes on startup
`Cable tester`	Tests network cables, USB cables for continuity
`Compressed air can`	Blows dust out of fans, heatsinks, keyboards
`Torch/flashlight`	Illuminates inside dark cases
`Parts organiser tray`	Holds screws and small parts during assembly

▸ 1.2 — Safety Rules in the Workshop

- Always wear an ESD wrist strap when handling components — static electricity kills chips
- Power off and unplug computer before opening case — never work on live equipment
- Hold circuit boards by edges — never touch chips or solder points
- Keep liquids away from equipment — even small drops cause short circuits
- Keep workspace clean and organised — loose screws can short motherboard
- Use correct screwdriver size — wrong size damages screw heads
- Store components in anti-static bags when not in use
- Never force connectors — if it does not fit, check orientation

▸ 1.3 — First Aid

Term / Concept	Definition / Explanation
`Electric shock`	Do NOT touch victim while connected to power — disconnect power first — call for help
`Burns`	Cool with running cold water for 10+ minutes — do not use ice or butter
`Eye injury`	Flush with clean water for 15+ minutes — seek medical attention
`Cuts`	Apply pressure with clean cloth — elevate if possible
`Fire`	Use correct extinguisher — evacuate if large — do not use water on electrical fires

▸ 1.4 — Fire Extinguisher Types

Term / Concept	Definition / Explanation
`Class A (Water/Foam)`	Wood, paper, textiles — NOT for electrical fires
`Class B (CO2/Dry powder)`	Flammable liquids — petrol, oil
`Class C (CO2/Dry powder)`	ELECTRICAL fires — computers use this type — CO2 does not damage equipment
`Class D`	Metal fires — magnesium etc.
`CO2 extinguisher`	Best for computer rooms — does not leave residue — does not damage equipment

> ⚠️ *NEVER use water or foam extinguishers on electrical fires — water conducts electricity and can cause electrocution.*

UNIT 2: COMPUTER SYSTEM USAGE & ACCESSORIES

Connecting, powering, and basic maintenance

▸ 2.1 — Connecting a Computer System

1. Place monitor at comfortable eye level distance
2. Connect VGA/HDMI/DisplayPort cable from monitor to PC
3. Connect keyboard and mouse (USB or PS/2)
4. Connect network cable (RJ-45 Ethernet) to NIC port
5. Connect power cables to monitor and PC — check voltage (220V Uganda)
6. Power on monitor first, then press PC power button

▸ 2.2 — Basic Maintenance Tasks

Term / Concept	Definition / Explanation
`Antivirus scan`	Run full system scan regularly — update definitions before scanning

`Disk cleanup`	Remove temporary files, empty Recycle Bin, clear browser cache
`Software updates`	Apply OS and application security updates regularly
`Defragmentation`	HDDs only — run monthly — SSDs should NOT be defragmented
`Backup`	Regular backups to external drive or cloud — follow 3-2-1 rule
`Dust removal`	Open case every 6 months — blow out dust with compressed air

▸ 2.3 — Energy Star Configuration

- Enable sleep/hibernate after inactivity — reduces power consumption
- Set monitor to turn off after 10-15 minutes idle
- Enable power management in BIOS
- Unplug devices not in use — standby still consumes power

UNIT 3: ELECTRONICS FUNDAMENTALS FOR MAINTENANCE

Electrostatics, colour codes, multimeters, component testing

▸ 3.1 — Electrostatic Discharge (ESD)

Static electricity builds up on your body. When you touch a component, it discharges through the component — potentially destroying it. Just 20 volts can damage a chip that runs on 3.3V or 5V. You may feel nothing, but the chip is damaged.

Term / Concept	Definition / Explanation
`ESD wrist strap`	Continuously drains static from body to ground — most important tool
`Anti-static bag`	Pink/silver metallic bags — store components in these when not installed
`Anti-static mat`	Work surface that drains static — connect wrist strap to mat
`Humidity`	Low humidity increases static — humid environments are safer
`ESD-safe clothing`	Special conductive clothing — used in professional clean rooms

▸ 3.2 — Resistor Colour Code

Resistors use coloured bands to indicate their value. The standard 4-band colour code:

Term / Concept	Definition / Explanation
`Black`	0
`Brown`	1
`Red`	2
`Orange`	3
`Yellow`	4

`Green`	5
`Blue`	6
`Violet`	7
`Grey`	8
`White`	9
`Gold (tolerance)`	±5%
`Silver (tolerance)`	±10%

> 💡 *Memory aid: B B ROY Great Britain Very Good Wife — Black Brown Red Orange Yellow Green Blue Violet Grey White*

▸ 3.3 — Using a Multimeter

Term / Concept	Definition / Explanation
`Voltage (V)`	Set to DC (for batteries/circuits) or AC (for mains) — place probes across component
`Current (A)`	Set to A range — place in SERIES with circuit (break circuit, insert meter)
`Resistance (Ω)`	Set to Ω range — POWER OFF — place across component
`Continuity`	Beeps when circuit is complete — tests cables and fuses
`Diode test`	Tests diodes — shows voltage drop (~0.6V for normal diode)

▸ 3.4 — Testing Basic Components

Term / Concept	Definition / Explanation
`Fuse`	Set multimeter to continuity — should beep if good — no beep = blown
`Switch`	Continuity test — should beep when switched ON, silent when OFF
`Cable/wire`	Continuity test — beep = wire is intact — no beep = broken
`Battery`	Voltage test DC — AA = ~1.5V, 9V battery = ~9V
`PSU`	Use 20-24 pin ATX tester or jump green wire to any black wire — fan should spin

UNIT 4: PRINTERS & SCANNERS

Types, operation, troubleshooting, maintenance

▸ 4.1 — Types of Printers

Term / Concept	Definition / Explanation

`Inkjet`	Sprays ink droplets — colour — home use — slow — cheap printer, expensive ink
`Laser`	Uses toner powder + heat — fast — office use — high quality text
`Dot-matrix`	Impact pins strike ribbon — very slow — used for multi-part forms, receipts
`Thermal`	Heat-sensitive paper — receipt printers, label printers — no ink/toner
`3D printer`	Builds 3D objects layer by layer — plastic filament or resin
`Plotter`	Moves pen over paper — engineering drawings — precise lines

▸ 4.2 — Printer Connectivity

Term / Concept	Definition / Explanation
`USB`	Most common — direct cable connection — plug and play
`Network (wired)`	RJ-45 Ethernet — shared printer on LAN — needs static IP
`WiFi`	Wireless network connection — convenient — connect via router
`Bluetooth`	Short range — mobile printing
`Parallel port (LPT)`	Old standard — 25-pin — mostly obsolete

▸ 4.3 — Printer Problems & Troubleshooting

Term / Concept	Definition / Explanation
`No output`	Check power, cable, driver, print queue — clear stuck jobs
`Poor print quality`	Low ink/toner, dirty print head, wrong paper — run head cleaning
`Paper jam`	Open all access doors, remove paper gently without tearing, check rollers
`Streaks on output`	Dirty drum (laser), clogged nozzle (inkjet) — clean or replace
`Printer offline`	Check connection, restart print spooler, check IP address (network printers)
`Slow printing`	Draft mode setting, network congestion, insufficient printer memory

▸ 4.4 — Scanner Types & Operation

Term / Concept	Definition / Explanation
`Flatbed scanner`	Glass surface, lid — highest quality — for photos and documents
`Sheet-fed scanner`	Paper fed through — faster, higher volume — cannot scan books
`Handheld scanner`	Portable — manual dragging — lower quality
`Drum scanner`	Professional — highest quality — used in publishing

`Pre-scan`	Quick low-resolution preview before final scan — adjust settings
`Resolution (DPI)`	Dots per inch — higher DPI = better quality but larger file — 300 DPI minimum for documents

UNIT 5: POWER PACKS & UPS

Power supply, UPS, troubleshooting power problems

▸ 5.1 — Computer Power Supply Unit (PSU)

Term / Concept	Definition / Explanation
`Function`	Converts AC mains (220-240V Uganda) to DC voltages used by components
`+12V rail`	Powers CPU, GPU, motors
`+5V rail`	Powers motherboard, older drives, USB
`+3.3V rail`	Powers RAM, modern components
`-12V rail`	Some serial ports
`Wattage`	Total power output capacity — 400W-1000W typical
`80 PLUS rating`	Efficiency rating — Bronze Silver Gold Platinum — higher = less waste heat

▸ 5.2 — UPS (Uninterruptible Power Supply)

Term / Concept	Definition / Explanation
`Purpose`	Provides battery backup during power outage — gives time to save and shutdown
`Offline/Standby UPS`	Switches to battery when mains fails — slight delay — cheapest
`Line-interactive UPS`	Regulates voltage fluctuations — good for Uganda with unstable power
`Online/Double-conversion`	Always on battery — cleanest power — most expensive
`VA rating`	Volt-Ampere — capacity — match to total equipment wattage
`Runtime`	How long battery lasts — depends on load vs capacity

▸ 5.3 — Power Problem Troubleshooting

Term / Concept	Definition / Explanation
`Computer won't turn on`	Check wall socket, check PSU switch at back, test PSU, check power button cable
`Random shutdowns`	Overheating (clean dust), failing PSU, power fluctuations
`Beeping on startup`	POST beep codes — short single beep = normal, multiple beeps

	= hardware error
`PSU failure signs`	Won't power on, burning smell, fan not spinning, random reboots
`PSU testing (multimeter)`	12V yellow wire, 5V red wire, 3.3V orange wire — test against black (ground)

UNIT 6: E-WASTE MANAGEMENT

Electronic waste — definition, categories, effects, management

▸ 6.1 — What is E-Waste?

E-waste (electronic waste) refers to discarded electrical or electronic devices. This includes obsolete computers, phones, TVs, printers, and any electronic device at end of life.

▸ 6.2 — Categories of E-Waste

Term / Concept	Definition / Explanation
`Large IT equipment`	Servers, mainframes, large printers, photocopiers
`Small IT equipment`	PCs, laptops, monitors, printers, scanners, keyboards, mice
`Telecom equipment`	Mobile phones, routers, switches, landlines
`Consumer electronics`	TVs, radios, cameras, game consoles, stereos
`Household appliances`	Microwaves, washing machines, refrigerators
`Lighting equipment`	Fluorescent tubes, LED strips, bulbs

▸ 6.3 — Effects of E-Waste

- Lead (in CRT monitors, solder) — neurotoxin — damages brain and nervous system
- Mercury (in LCD backlights) — accumulates in food chain — damages kidneys and brain
- Cadmium (in rechargeable batteries) — carcinogen — damages kidneys
- Brominated flame retardants — toxic gases when burned — lung damage
- Soil contamination — chemicals leach into ground — damages crops and groundwater
- Air pollution — burning e-waste releases toxic fumes — respiratory problems
- Water pollution — chemicals reach rivers and lakes — affects aquatic life

▸ 6.4 — E-Waste Management

Term / Concept	Definition / Explanation
`Personal level`	Reduce — buy durable equipment. Reuse — donate working equipment. Recycle — give to certified recyclers
`Institutional level`	Asset register for IT equipment, data wiping before disposal, certified recycling contracts
`Government level`	E-waste legislation, certified recycling centres, import regulations

	on old equipment
`E-waste cycle`	Collection → Sorting → Disassembly → Material recovery → Safe disposal of hazardous parts

> 📝 *Exam: Know the 3Rs of E-Waste: Reduce, Reuse, Recycle. Know at least 3 hazardous materials in e-waste and their health effects.*